

Open-Ended Self-Guided Workbook

Core1B stays close to taught mathematics while support fades. Every task begins open-ended and offers staged self-help plus a numeric/graphical bottom check.

This sample demonstrates:

- Open question before explanation.
- Self-help: small clue -> bigger clue -> how do I start?
- Detailed answer on the same page bottom.
- No live branching and no learner-state claim.

Sample topic

A point on an axis equidistant from two fixed points

Generated review fixture - not an official exam source

1. Two methods

Attempt first. Use only as much help as needed. Detailed answer stays at the bottom.

TRY IT FIRST

$A=(1,2)$, $B=(5,2)$. P lies on x -axis and $PA=PB$. Find P using symmetry and the general model.

SMALL CLUE

Which coordinate of P is fixed by the axis?

BIGGER CLUE

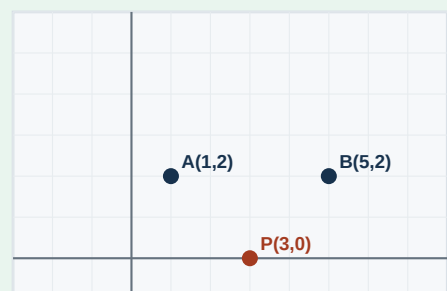
Use equality of squared distances.

HOW DO I START?

Write $P=(x,0)$ or $P=(0,y)$, then form $PA^2=PB^2$.

CHECK AFTER YOU TRY - DETAILED ANSWER

$P=(x,0)$. Symmetry gives $x=(1+5)/2=3$. General check:
 $(x-1)^2+4=(x-5)^2+4 \rightarrow x=3$. Therefore $P=(3,0)$.



2. Reject an unsafe shortcut

Attempt first. Use only as much help as needed. Detailed answer stays at the bottom.

TRY IT FIRST

$A=(-1,4)$, $B=(7,2)$. A student claims $P=(3,0)$ by averaging x-coordinates. Test the claim and find the real P.

SMALL CLUE

Which coordinate of P is fixed by the axis?

BIGGER CLUE

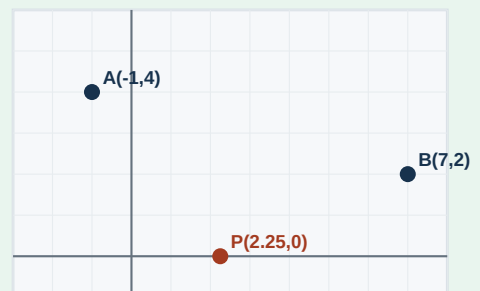
Use equality of squared distances.

HOW DO I START?

Write $P=(x,0)$ or $P=(0,y)$, then form $PA^2=PB^2$.

CHECK AFTER YOU TRY - DETAILED ANSWER

At $(3,0)$, $PA^2=32$ while $PB^2=20$, so the guess fails. Solve $(x+1)^2+16=(x-7)^2+4 \rightarrow 16x=36 \rightarrow x=9/4$. $P=(9/4,0)$.



3. Axis changes

Attempt first. Use only as much help as needed. Detailed answer stays at the bottom.

TRY IT FIRST

$A=(2,-1)$, $B=(2,7)$. P lies on y -axis and $PA=PB$. Find P and state what changed.

SMALL CLUE

Which coordinate of P is fixed by the axis?

BIGGER CLUE

Use equality of squared distances.

HOW DO I START?

Write $P=(x,0)$ or $P=(0,y)$, then form $PA^2=PB^2$.

CHECK AFTER YOU TRY - DETAILED ANSWER

Now $P=(0,y)$. $4+(y+1)^2=4+(y-7)^2 \rightarrow y=3$. Therefore $P=(0,3)$. The axis changed; the equal-distance invariant did not.

